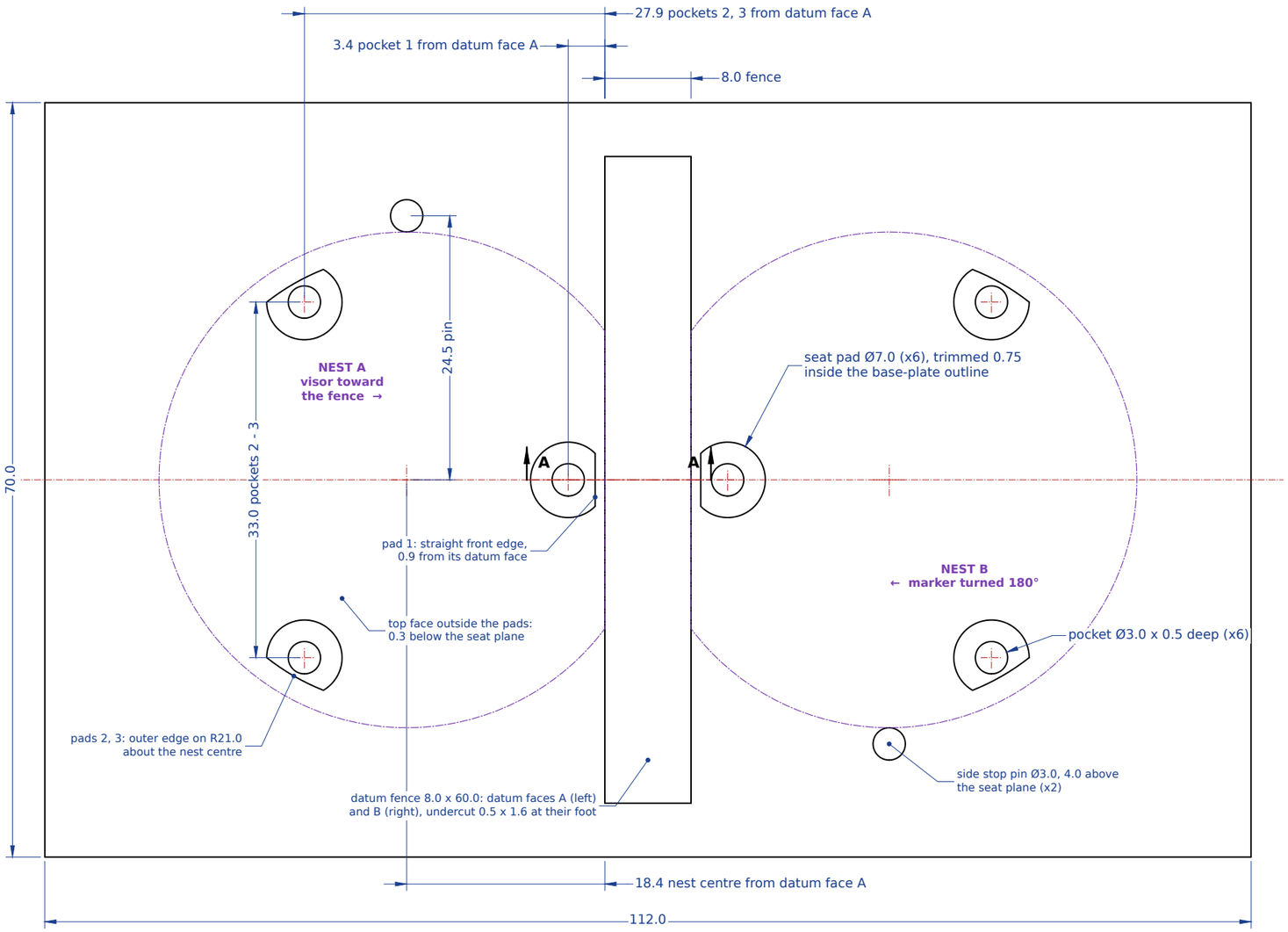
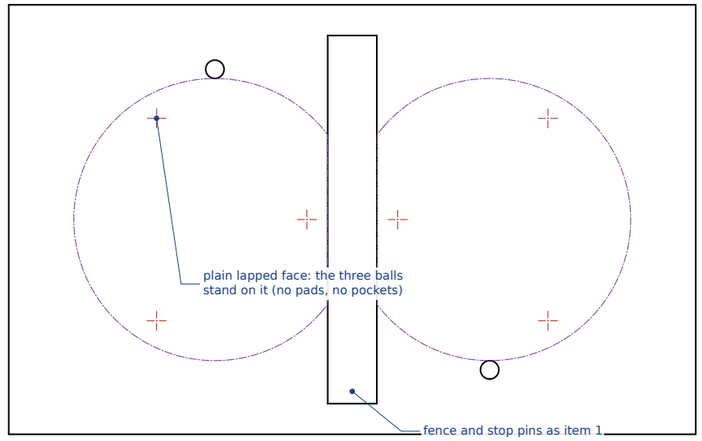
ITEM 1 - BASE-FACE SEAT PLATE, plan 2:1 (the marker is shown dash-dot in both nests)



ITEM 2 - BALL SEAT PLATE, plan 1:1



Requirement	Value
blank, both items	70.0 x 112.0 x 12.0 (seat plane to underside)
item 1: the six pad tops in one plane	0.001 (lapped together); Ra 0.1
item 2: flatness of the top face	0.001 over each nest, 0.002 overall; Ra 0.1
fence: faces A and B square to the seat plane, parallel	0.01
pocket (item 1)	Ø3.0 +0.1 / 0 x 0.5 +0.1 / 0, flat bottom
seat pad (item 1)	Ø7.0, 0.3 +-0.05 proud of the top face
seat pad trim (item 1)	pads 2, 3: R21.0 about the nest centre; pad 1: straight, 0.9 from its datum face
pocket and pin positions (table on sheet 2)	+-0.05 from the datum face and the centre line
underside	ground; parallelism to the seat plane not functional

NOTES

1. Fixtures for the factory level zero on TWO SEATS (DESIGN_SPEC section 6, E11). Item 1: the marker rests on its BASE FACE (as it floats on turf): three annular seat pads touch the FR-4 face next to the three feet and each ball hangs free in its pocket. Item 2: the marker stands on its three BALLS on a plain face.
2. POCKET: Ø3.0 clears the ball cap (Ø0.65 at the seat plane), the mask relief (Ø1.25) and a foot up to 0.6 off its pocket centre; 0.5 deep is 4.2 x the largest protrusion (0.12). Rim broken 0.05 max, no burr above the pad.
3. PADS, not the whole face: a legend print stands about 0.02 proud of the mask and the tub rim is 0.25 above the base plane. The base plate keeps its legend out of Ø8.5 round each foot (GL1-011 sheet 2).
4. TWO NESTS: read in nest A, lift, turn the marker 180°, read in nest B. Both nests lie in one lapped plane, so the tilt of the plate and of the table cancels; no parallelism of the plate is needed.
5. LOCATION: push the visor flat against the fence (contact above the undercut, z 1.6 to 4.0) and the round wall against the stop pin.
6. NO CONTACTS, no clamp, no cable: a spring pin pushes five times harder than the marker weighs. The zero is read with the marker free on the plate.
7. Clean plate and marker before every read (10 µm of dust is 0.4 mrad).
8. The six-face fixture (a cube nest with spring pins) is NOT on this drawing: not yet designed (README, fixtures).
9. 0 AND 180 DEGREES ONLY: the fence takes the visor flat, so these plates give the two-position reversal. A four-position zero (0, 90, 180, 270; planned for production) needs a location on the round wall instead: not designed.
10. Seat pads end 0.75 inside the base-plate outline (R21.75, flat at 18.25 from the nest centre): pads 2, 3 on R21.0, pad 1 on a line 0.9 from its datum face.
11. Item 3, the ball gauge plate of assembly step 2: sheet 2.
12. PROVISIONAL - not released. Nothing has been made. Checked by check 18.

TITLE

GL1-050 Calibration seat plates and ball gauge plate (fixtures)

PROJECT / PART

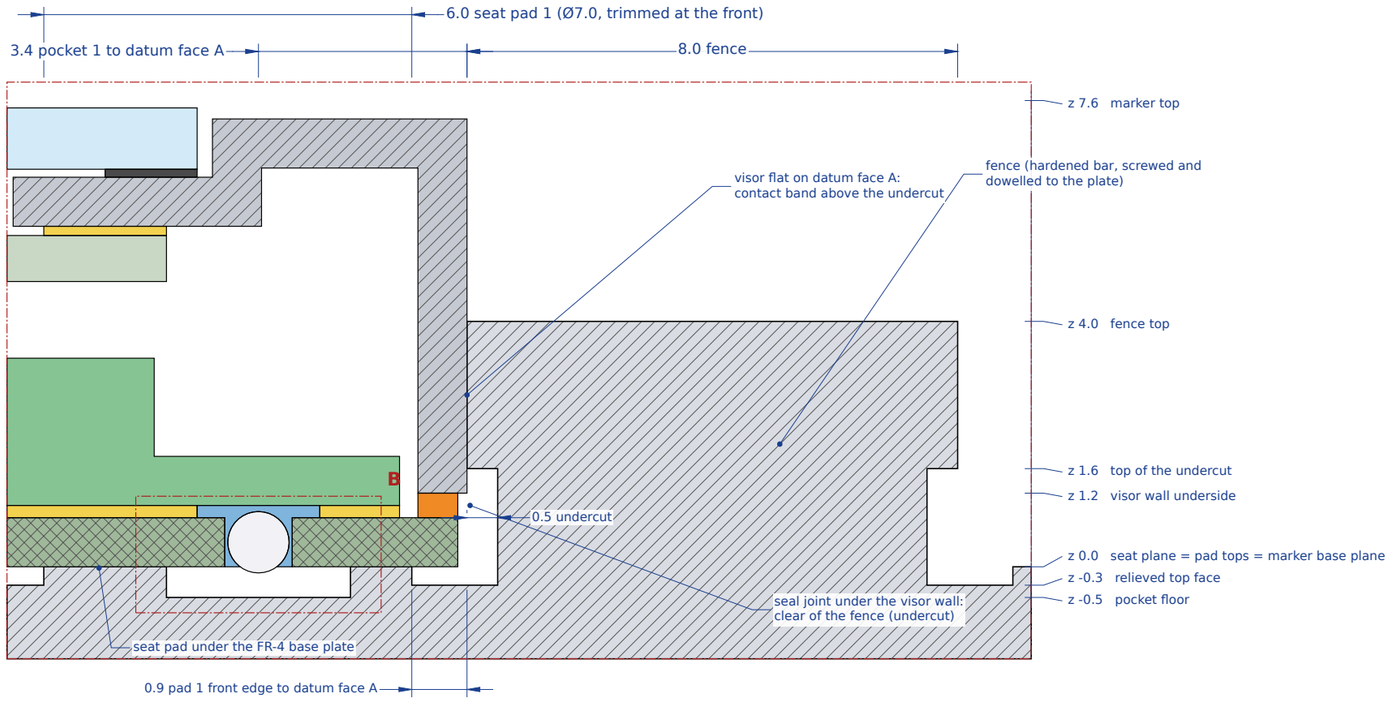
GreenLine GL-1 electronic ball marker - fixtures: 1 base-face seat plate, 2 ball seat plate, 3 ball gauge plate

MATERIAL / FINISH	PROCESS
tool steel, hardened 58-62 HRC, stabilised; seat faces lapped	ground, pockets milled, seat faces lapped; fence and pins fitted
GENERAL TOLERANCE	SCALE (A3) / UNITS
+-0.1 unless stated; seat faces: see table	2:1 / 1:1 mm

STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1.2 / D-EVT1	1 of 2 2026-11-13

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

SECTION A-A (plane x = 0 of nest A): marker on its base face in item 1, visor flat on datum face A 10:1



POSITIONS (mm; plate centre line = marker x = 0; nest B = nest A turned 180°)

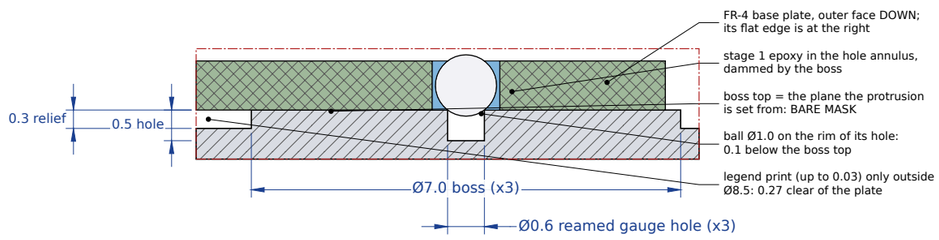
Feature	Nest	across the centre line (marker x)	from its datum face	datum face
pocket 1 / seat pad 1	A	0.0	3.4	A
pocket 2 / seat pad 2	A	-16.5	27.9	A
pocket 3 / seat pad 3	A	16.5	27.9	A
stop pin Ø3.0	A	-24.5	18.4	A
pocket 1 / seat pad 1	B	0.0	3.4	B
pocket 2 / seat pad 2	B	16.5	27.9	B
pocket 3 / seat pad 3	B	-16.5	27.9	B
stop pin Ø3.0	B	24.5	18.4	B
nest centre (marker centre)	A, B	0.0	18.4	A, B

NOTES

1. Pocket positions are the ball-foot positions of the marker (YAML base_plate.feet): (0.0, 15.0), (-16.5, -9.5), (16.5, -9.5) from the marker centre; the visor flat is at y = 18.4, the round wall at R23.0.
2. Section A-A is cut through nest A with the marker seated (assembly solids of this build). The fence touches the visor flat between the top of its undercut and its top; the window recesses start higher (z 4.2).
3. Item 2 in the same section: no pads and no pockets, the balls stand on the face, the base plate is 0.1 above it.

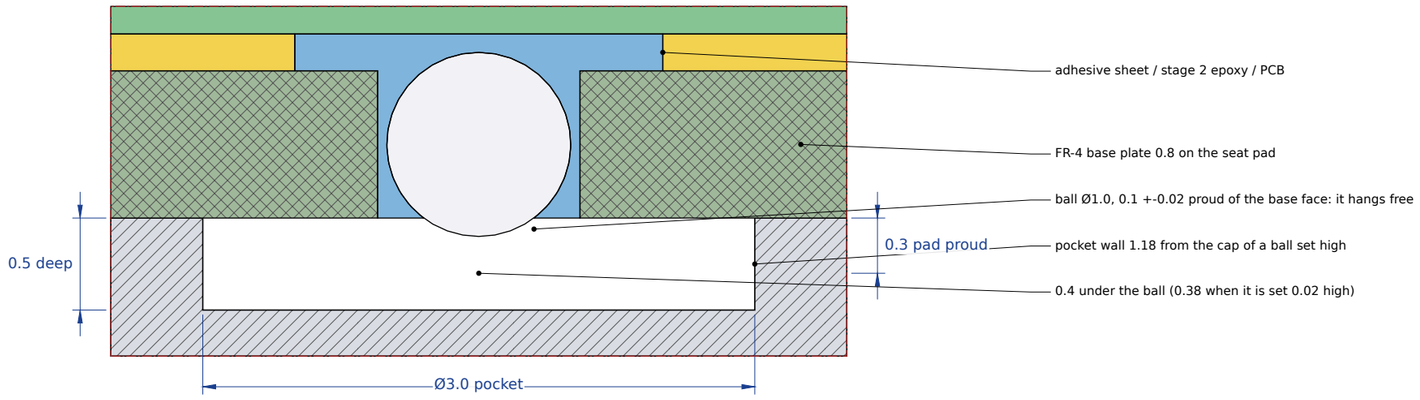
ITEM 3 - BALL GAUGE PLATE (assembly step 2): section x = 0 through boss 1 10:1

the base plate lies on it OUTER FACE DOWN (shown at step 2: ball and stage 1 epoxy only)



4. ITEM 3: blank 52.0 x 50.0 x 10.0; three bosses Ø7.0, 0.3 proud, tops ground in one plane within 0.005; in each a reamed hole Ø0.6 +0.005 / 0 x 0.5, sharp burr-free rim, at the pocket positions of the table (+-0.02). The base plate rests on the bosses only (inside its legend-free zones); clamp: three annular feet over them.

DETAIL B - ball foot 1 in its pocket 30:1



Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE: **GL1-050 Calibration seat plates and ball gauge plate (fixtures)**		
PROJECT / PART: GreenLine GL-1 electronic ball marker - fixtures: 1 base-face seat plate, 2 ball seat plate, 3 ball gauge plate		
MATERIAL / FINISH: tool steel, hardened 58-62 HRC, stabilised; seat faces lapped	PROCESS: ground, pockets milled, seat faces lapped; fence and pins fitted	
GENERAL TOLERANCE: +-0.1 unless stated; seat faces: see table	SCALE (A3) / UNITS: 10:1 / 30:1 mm	
STATUS: **PROVISIONAL - not released**	DWG REV / PARAMS: D-EVT1.2 / D-EVT1	SHEET / DATE: 2 of 2 2026-11-13